財政學 HW3

1. (Hyman 3.3) The EPA wants to reduce emissions of sulfur dioxides from electric power-generating plants by 20 percent during the next year. To achieve this goal, compensate receptors of external damage in cases where there are few emitters and many receptors?
the EPA will require each power-generating plant in the nation to reduce emissions by 100 tons per year. Suppose five power plants emit sulfur dioxides and serve a given metropolitan area. The following table shows the cost per ton of reducing emissions for each of the five plants:

COST PER TON OF PLANT	EMISSIONS REDUCTION ($)
1	600
2	500
3	500
4	400
5	200

Assuming that the cost per ton of emissions reduction is constant and that the improvement in the air for the metropolitan area is the same no matter which plant reduces emissions, calculate the following:
a. Cost of meeting EPA regulations.
b. Least-cost method of achieving the EPA goal of reducing emissions of sulfur dioxides from power plants in the metropolitan area

ANS
a. The cost of meeting the standards is obtained by multiplying cost per ton for each plant and summing the results, which gives $220,000 per year.
600(第一間工廠減少每噸排放量所花費的成本)*100(噸) + 500*100 + 500*100 +400*100 + 200*100 = 220,000
b. The least-cost method would be to have Plant 5 reduce its emissions by 500 tons per year, which would cost only $100,000 per year.
200(第五間工廠減少每噸排放量所花費的成本)*500(噸) = 100,000

2. (Hyman 3.4) Instead of using regulations to achieve the 20 percent reduction in emissions discussed in the preceding problem, suppose the EPA requires each of the

five emitters to pay a fee of $450 for each ton of sulfur dioxide it dumps in the air during the year.
Use the data from the table for problem 1 to predict which companies will purchase pollution rights, total cost of achieving the reduction in sulfur dioxide emissions, and revenue generated from the sale of pollution rights in the area.

ANS
At a $450 per ton charge, Plants 4 and 5 would cut back emissions at a total cost of $60,000(=400*100 + 200*100). Plants 1, 2, and 3 would find it cheaper to pay the $450 fee, rather than cut back emissions. The total revenue generated by the purchase of the pollution rights by these firms for the 300-ton reduction in annual emissions would therefore be $135,000(= 450*300).

3. (Rosen 5.7) For each of the following situations, is the Coase Theorem applicable? Why or why not?

a. A farmer who grows organic corn is at risk of having his corp contaminated by genetically modified corn grown by his neighbors.

b. In Silveiras, Brazil, queen ants are regarded as a delicacy, but recently the ants haul has been dwindling because of pesticides used on eucalyptus trees that are planted to produce cellulose for paper and other products [Barrionuevo and Demit, 2011]

c. Following a military coup in Madagascar, a weak president was installed who was unable to shop the selling of Rosewood trees to China that were illegally cut down form the national parks.

d. User of the Internet generally incur a zero incremental cost for transmitting information. As a consequence, congestion occurs, and users are frustrated by delays.

ANS
a. It is very likely that the farmer could negotiate with the neighbors, provided property rights are clearly defined. The Coase Theorem is therefore applicable.
b. It is unlikely that negotiation could result in an efficient outcome in this case. It is likely that there are a great number of farmers involved in both harvesting ants and growing trees, making negotiation very difficult.
c. Property rights are not being enforced, making negotiation through the Coase Theorem impossible.

d. There are too many people involved for private negotiation.

4. (Rosen 5.9) In India, a drug used to treat sick cows is leading to the death of many vultures that feed off of dead cattle. Before the decrease in the number of vultures, they sometimes used to smash into the engines of jets taking off form New Delhi's airports, posing a serious threat to air travelers. However, the decline of the vulture population has led to a sharp increase in the populations of rats and feral dogs, which are now the main scavengers of rotting meat [Gentleman, 2006, p. A4]. There have been calls for a ban on the drug used to treat the cows.
Identify the externalities that are present in this situation. Comment on the efficiency of banning the drug. How would you design an incentive-based regulation to attain an efficient outcome?

ANS

The use of the drug to treat sick cows leads to a positive externality (the benefit enjoyed by air travelers) as well as a negative externality (the costs created by a larger number of rats and feral dogs). Banning the drug might raise or lower efficiency, depending on whether the positive externality is larger or whether the negative externality is larger. There are many ways to design incentive-based regulations. Policymakers could determine the efficient level of drug usage and then either allocate or sell the right to use the drug for sick cows.

5. (Rosen 5.11) American suburbs are expanding to more rural areas at the same time as a pig farms are expanding in size [*Economist*, 2007d, p. 36]. The smells emanating from the massive amounts of pig manure adversely affect property values. Imagine that the Little Pigs (LP) hog farm is situated near 100 houses. The following table shows, for each level of LP's output, the marginal cost (MC) of a hog, the marginal benefit (MB) to LP, and the marginal damage (MD) done to property values:

Output	MC	MB	MD
1	400	1600	400
2	800	1600	800
3	1200	1600	900
4	1600	1600	1000
5	3200	1600	1200
6	6400	1600	1400

a. How many hogs does LP produce?
b. What is the efficient number of hogs?

c. Suppose the owner of LP can reduce the marginal damages of hog smells by two-thirds by modifying the hogs' diet. The modified diet increases the marginal cost of each hog by $100. What is the efficient number of hogs?

ANS

a. When the Little Pigs hog farm produces on its own, it sets marginal benefit equal to marginal cost. This occurs at 4 units.

b. The efficient number of hogs sets marginal benefit equal to marginal social cost, which is the sum of MC and MD. At 2 units, MB=MSC=1600.

c. The efficient number of hogs sets marginal benefit equal to marginal social costs. At 3 hogs, MB=MSC=1600.

6. 閱讀老師稅評「政府關門？總預算案僵局的真相看過來」，文章中有非常多 法條值得留意，能幫助同學們更了解總預算案的架構，請根據題目敘述找出相對應的法條。(共有7小題，第a小題為範例)

a.根據預算法，總預算案應於會計年度開始一個月前由立法院完成審議（議決），並於會計年度開始15日前由總統公布。

答:預算法第51條

總預算案應於會計年度開始一個月前由立法院議決，並於會計年度開始十五日前由總統公布之；預算中有應守秘密之部分，不予公布。

b.根據立法院職權行使法，預算案應經三讀會議決。

答: 立法院職權行使法第7條

立法院依憲法第六十三條規定所議決之議案，除法律案、預算案應經三讀會議決外，其餘均經二讀會議決之。

c.根據預算法，總預算案未能於期限內完成議決，各機關預算之執行，按「馬照跑、舞照跳」的原則處理；簡單來說，「該收的稅費照收」、「該發的金錢照 發」、「延續中計畫照做」、「必須借的錢照借」，是以絕對不會出現政府關門的情形。在「馬照跑、舞照跳」原則下，唯有「新增計畫」與「新興資本支出」，須俟預算案完成審議程序後，始得動支。

答: 預算法第54條

總預算案之審議，如不能依第五十一條期限完成時，各機關預算之執行，依下列規定為之：

一、收入部分暫依上年度標準及實際發生數，覈實收入。

二、支出部分：

（一）新興資本支出及新增計畫，須俟本年度預算完成審議程序後始得動支。但依第八十八條規定辦理或經立法院同意者，不在此限。

（二）前目以外計畫得依已獲授權之原訂計畫或上年度執行數，覈實動支。

三、履行其他法定義務收支。

四、因應前三款收支調度需要之債務舉借，覈實辦理。

d.根據預算法，新增計畫與新興資本支出，在預算案編列的額度內，只要立法院同意，仍可執行！

答: 預算法第54條

總預算案之審議，如不能依第五十一條期限完成時，各機關預算之執行，依下列規定為之：

一、收入部分暫依上年度標準及實際發生數，覈實收入。

二、支出部分：

（一）新興資本支出及新增計畫，須俟本年度預算完成審議程序後始得動支。但依第八十八條規定辦理或經立法院同意者，不在此限。

（二）前目以外計畫得依已獲授權之原訂計畫或上年度執行數，覈實動支。

三、履行其他法定義務收支。

四、因應前三款收支調度需要之債務舉借，覈實辦理。

e.行政院訂有中央政府總預算案未能依限完成時之執行補充規定的行政規章，正是為了避免總預算案之審議，因故全部或部分未能通過，而使政府施政中輟。

答: 中央政府總預算案未能依限完成時之執行補充規定

中華民國104年12月23日 行政院院授主預字第 1040102760D 號函修正

f. 根據憲法，立法院會期，每年兩次。

答: 中華民國憲法第68條

立法院會期，每年兩次，自行集會，第一次自二月至五月底，第二次自九月至十二月底，必要時得延長之。

g.根據憲法增修條文，每屆立委任期四年，故共有八個會期。

答: 中華民國憲法增修條文第4條

立法院立法委員自第七屆起一百一十三人，任期四年，連選得連任，於每屆任滿前三個月內，依左列規定選出之，不受憲法第六十四條及第六十五條之限制：

一、自由地區直轄市、縣市七十三人。每縣市至少一人。

二、自由地區平地原住民及山地原住民各三人。

三、全國不分區及僑居國外國民共三十四人。